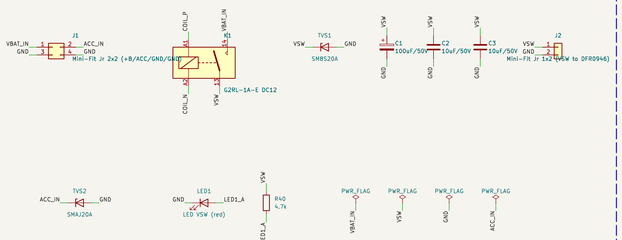
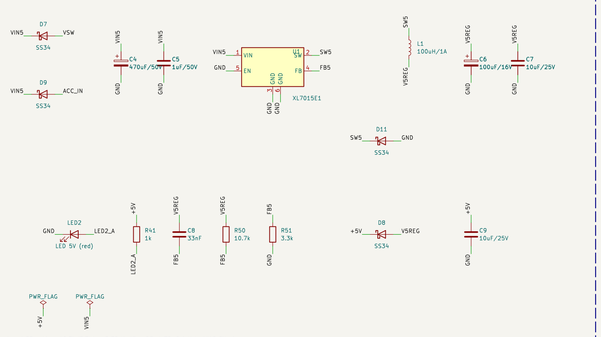
1. POWER INPUT / MAIN SWITCH (7 A path: J1 -> K1 -> J2)



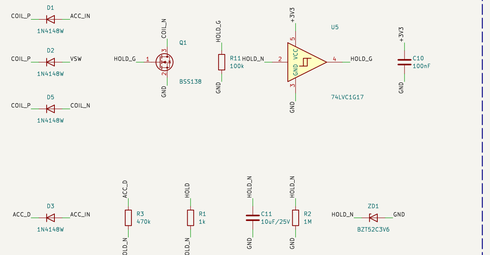
K1 contact 13–14 carries up to 7 A (PC 60 W at 9 V crank). F1 7.5 A / F2 1 A blade fuses are in the harness.
TVS1 after relay: zero leakage when off. Clamp 32.4 V < DFR0946 35 V max.

2. 5 V AUX SUPPLY (XL7015, Vout 5.30 V)



Vout = 1.25 × (1 + 10.7k/3.3k) = 5.30 V. After D8: 4.75–5.1 V to XIAO 5V pin.
Keep C5–U1–L1–C6 loop tight. EN tied to GND (enabled).

3. RELAY HOLD (ACC hold + MCU override + Schmitt)



ACC on & MCU Hi-Z -> ON (hardware start). MCU drives HOLD low -> OFF (1k beats 470k).
ACC off & MCU Hi-Z -> OFF after ~8 s (4.6–17 s, 10uF + 1M + ZD1 leak. Schmitt VT-: SPICE tbd). XIAO removed (no 3V3) -> OFF.

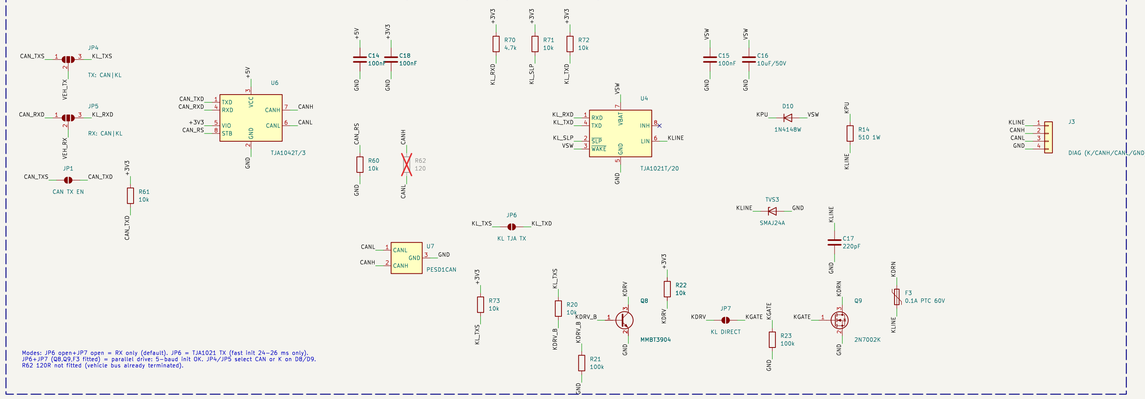
4. MCU: Seeed XIAO ESP32S3 (socketed)



ACC sense (ratio 7.91, 47k/6.8k)

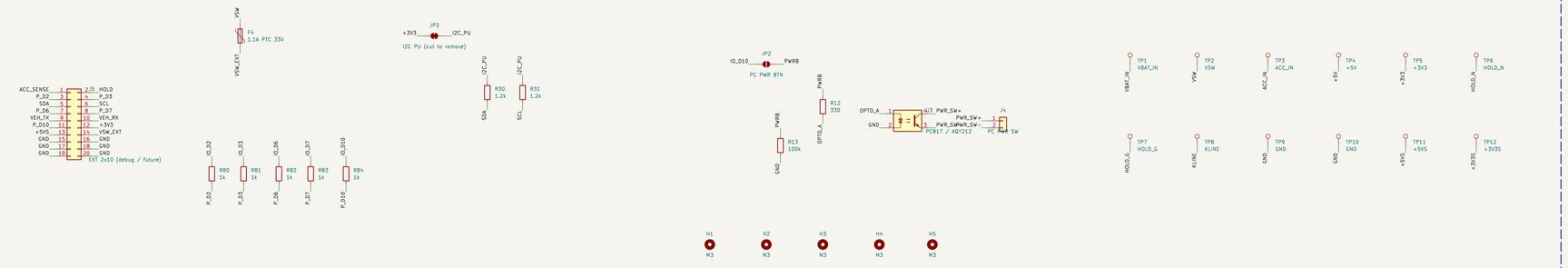
D6/D7 (GPIO43/44) carry UART0 boot log: avoid as outputs needing a quiet boot.
PC USB goes to XIAO USB-C through the separate VBUS-block adapter.

5. VEHICLE BUS: CAN (TJA1042T/3, bus +-58V, VCC 5V / VIO 3.3V, STB=L normal) / K-LINE (TJA1021) - default RX-only

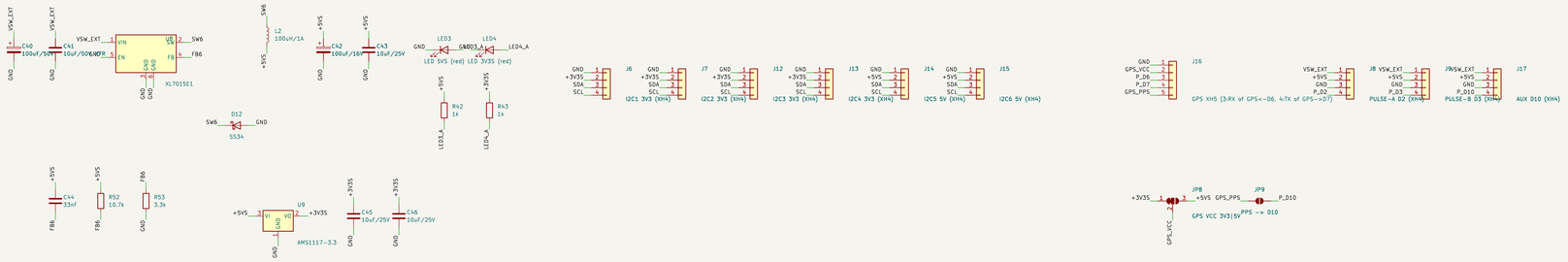


Modes: JP6 open+JP7 open = RX only (default). JP6 = TJA1021 TX (fast init 24–26 ms only).
JP6+JP7 (Q8,Q9,F3 fitted) = parallel driver. 5-baud init OK. JP4/JP5 select CAN or K on D8/D9.
R62 120R not fitted (vehicle bus already terminated).

6. EXPANSION HEADER / I2C PULL-UP / PC POWER BUTTON / TEST POINTS / MECHANICAL



7. SENSOR SUPPLY (5.3 V / 3.3 V from VSW_EXT) + PORTS (all JST XH)



Sensor rails are separate from the XIAO supply: +5VS ~0.7 A (XL7015). +3V3S ~0.4 A (AMS1117, 1 W).
Powered only while the relay is on -> zero dark current is kept.

I2C1–6 share one bus (D4/D5, 400 kHz, 1.2k pull-up; module pull-ups in total >= 5k). I2C5/6: 5 V power, 3.3 V logic.
PULSE-A/B and AUX: 1 12V 2 5V 3 GND 4 IO (same as PULSE-IN adapter). D10 is shared by AUX, GPS PPS (JP9) and the PC power button (JP2): use one.
RB0–RB4 1k: 1-Wire pull-up >= 4.7k, GPS UART <= 460800 bps.

OpenCruise
Sheet: /
File: pwctrl.kicad_sch
Title: OpenCruise PWR-CTRL main board
Size: A1 | Date: 2026-11-09 | Rev: 2.0
KiCad E.D.A. 9.0.9 | Id: 1/1